Can Five Seeds Tell Embedding Initializers Apart? A Controlled Study of Vocabulary Extension for Amharic

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Abstract

Extending a pretrained encoder's vocabulary to a new language requires an embedding for every added token, and several initialization strategies have been proposed to supply one. We ask whether the benchmarks commonly used for Amharic can distinguish between them. Under a single fixed protocol—one configuration, one set of rebuilt data splits, one untuned fine-tuning recipe, shared seeds—we compare morpheme-grounded initialization, overlap-based initialization, and no vocabulary extension at all, across text classification, named entity recognition and extractive question answering. Across six paired comparisons, every 95% confidence interval contains zero. On all three tasks the seed-to-seed variation exceeds the entire spread between the three systems, and on text classification the smallest difference five seeds could detect at conventional power is 19.8 macro-F1 points against an observed between-system spread of 9.0. A further property of the setup bounds what any such comparison can show: 6 of 60 runs never fit their training data and settle into degenerate predictors, distributed across all three systems and identifiable from training loss alone. We also record that a favourable comparison reported under a superseded protocol reverses under this one, and that the two protocols differ in test set, training set and class prior together, so the shift cannot be attributed to any single cause. We report means and intervals and claim no ordering. This paper replaces a retracted predecessor (Teklehaymanot et al., 2026); all code, data manifests, per-run records and the script that generates every number are released.

1 Introduction

A multilingual encoder adapted to a new language by vocabulary extension needs a vector for each token it has never seen. Random initialization discards what is known about the token's form. FOCUS (Dobler and de Melo, 2023) builds each new embedding from tokens it overlaps with in an auxiliary space. A morpheme-grounded alternative decomposes the token and averages over its morphemes, which is appealing for Amharic, where affixation is heavy and a frequency-driven tokenizer fragments surface forms arbitrarily.

Papers in this area report differences of one to several points between such strategies, with standard deviations below one point, and conclude in favour of one of them. This paper asks a prior question: at the sample sizes conventionally used, can these benchmarks support such a conclusion at all?

We answer it for Amharic by running three systems under one protocol and examining the variance structure rather than only the means. The answer is that they cannot. We also measure a fine-tuning instability that costs 10% of runs, and we document a favourable comparison reported under a superseded protocol that reverses under this one.

Relation to a retracted paper. An earlier paper by these authors described the morpheme-grounded method and reported a results table for which no archived experimental record exists, a projection matrix that was never specified, and a dataset that was not constructed. That paper (Teklehaymanot et al., 2026) is retracted. The present paper replaces it. It shares no text and no results with it. All data splits were rebuilt from their original sources and are documented in §3 and Appendix A, which maps every number reported here to the job identifier, seed and result file it came from.

Contributions.

1. A controlled three-way comparison on Amharic—60 runs, one configuration, shared seeds—in which no paired confidence interval excludes zero (§4).
2. A measurement of why: seed variation exceeds between-system spread on every task, and the minimum detectable difference at five

seeds is an order of magnitude larger than the effects under discussion (§5).

3. A characterization of a fine-tuning failure mode affecting 6 of 60 runs, separable from the method under test using training loss alone (§7).

2 Systems

XLM-R. Off-the-shelf XLM-R base (Conneau et al., 2020), fine-tuned directly on each task. No vocabulary extension, no adaptive pretraining, no auxiliary embedding space. This is the far control: it bounds what the rest of the machinery is worth.

FOCUS. Vocabulary extension with FOCUS initialization (Dobler and de Melo, 2023), followed by language-adaptive pretraining. Each new embedding is a weighted combination of overlapping tokens in a FastText space, projected into the model's space. This is the near control: it consumes the same auxiliary space and the same projection as the morpheme-grounded system, and differs only in not using morpheme segmentation.

LGSE. The morpheme-grounded initializer. A new token t with decomposition $m_1 \dots m_k$ receives

$$e_{\text{new}} = \frac{1}{k} \sum_{j=1}^{k} W e_{m_j}, \tag{1}$$

where e_{m_j} is a FastText vector and $W \in \mathbb{R}^{d \times d}$ aligns the two spaces. Tokens without a decomposition fall back to a whole-token vector and then to character n-grams. During adaptive pretraining a term $\lambda \|e_{\text{new}} - \mu\|^2$ penalizes drift from the initialization.

Two quantities the method does not fix. Equation 1 requires W, and the regularizer requires λ. Neither is determined by the method as described, and the retracted paper specified neither. We therefore chose both, documented the choice, and held it fixed for every run: W is obtained by orthogonal Procrustes, $W^\star = \arg\min_{W^\top W = I} \|WX - Y\|_F$, on 5,910 surface anchors shared between the Amharic FastText model and XLM-R (residual Frobenius norm 423.0 against 633.0 unaligned), and $\lambda = 1.0$. These are our choices. A reader should treat the LGSE column as one faithful instantiation rather than the only one.

We note one structural consequence. The initializer writes We_{m_j} into the embedding matrix and the regularizer anchors to a constant, so nothing downstream is a function of W once initialization is complete. W is a fixed preprocessing transform here, not a learned parameter.

3 Setup

One protocol. All 60 runs share a single configuration file, verified by checksum: every run records the same `config_sha256`. Fine-tuning uses a constant learning rate of 5×10^{-5}, no warmup, batch size 32, 10 epochs, and checkpoint selection by evaluation loss. This recipe is identical for all three systems and was not tuned for any of them; the selection metric is task-agnostic by construction. Appendix A records the checksum and the full recipe.

Data. For NER we take the MasakhaNER splits (Adelani et al., 2021) unchanged, and for question answering the AmQA splits (Taffa et al., 2024) unchanged. Text classification uses a 2,499-row annotated educational-quality corpus, reduced to 2,340 after removing 2 spreadsheet-error rows and 157 exact duplicates, split 1872/233/235 by a stratified procedure at seed 42 and verified leakage-free. The first author led that annotation. Each split carries a manifest with source, raw checksum and split seed.

Metrics. F1 for NER and QA. For text classification we report macro-F1 over the five classes present in the test split as the primary metric, with accuracy secondary. The majority class holds 66.4% of the test set, so accuracy is dominated by it: the majority-class baseline is 66.38 accuracy against 15.96 macro-F1. We also report the macro-F1 value recorded in the run files, which averages over six classes including one with no test instances and is therefore capped at five sixths of the attainable maximum.

Seeds. 42–46 for every cell, extended to 42–51 on NER for all three systems after the failure mode of §7 appeared there.

Scope. Amharic only, three systems. Two further systems from the retracted design—adaptive pretraining alone, and extension with random initialization—were not run under this protocol, and §10 explains why results from an earlier batch cannot substitute for them.

Task		LGSE	FOCUS	XLM-R
TC	macro-F1	45.61	54.56	45.59
	sd	12.80	2.59	15.04
	accuracy	78.30	81.62	78.98
	n / fail	5 / 2	5 / 0	5 / 2
NER	F1	61.44	60.10	67.05
	sd	21.64	21.30	2.55
	n / fail	10 / 1	10 / 1	10 / 0
QA	F1	58.23	59.41	59.54
	sd	1.40	1.51	2.14
	n / fail	5 / 0	5 / 0	5 / 0

Table 1: All 60 runs, one protocol. TC macro-F1 is over the five classes present in test; the majority-class baseline is 15.96 macro-F1 and 66.38 accuracy. *fail* counts runs that never fit their training data (§7); they are retained here. No ordering is claimed: see Table 2.

Task	vs	n	Δ	95% CI
TC	FOCUS	5	-8.95	$[-23.74, +5.84]$
	XLM-R	5	$+0.03$	$[-20.53, +20.58]$
NER	FOCUS	10	$+1.34$	$[-21.61, +24.29]$
	XLM-R	10	-5.61	$[-22.03, +10.80]$
QA	FOCUS	5	-1.17	$[-2.91, +0.56]$
	XLM-R	5	-1.30	$[-3.12, +0.51]$

Table 2: Paired differences, LGSE minus comparator, on matched seeds. Positive favours LGSE. **Every interval contains zero.** Exact Wilcoxon agrees (0.188–0.812); at these sample sizes the smallest attainable two-sided exact p is 0.062 at $n = 5$ and 0.125 at $n = 4$, so four of these six rows could not have reached $p < 0.05$ at any effect size, and we do not report p-values as evidence.

4 Results

Table 1 gives the three systems under the common protocol and Table 2 the paired differences.

No interval excludes zero. The largest observed difference is -8.95 macro-F1 on text classification against FOCUS, with an interval spanning -23.74 to $+5.84$. The smallest is $+0.03$ against XLM-R, an interval 41 points wide. Nothing in Table 2 supports an ordering of the three systems.

Win counts are more informative than the means, and still insufficient. LGSE loses 4 of 5 QA seeds to each comparator and wins 0 of 5 text-classification seeds against FOCUS, which is a consistent direction. But five paired observations cannot certify a direction at conventional thresholds, and §5 shows the margin by which they fall short.

Task	Spread	Mean sd	Ratio
TC	8.98	10.14	0.89
NER	6.95	15.16	0.46
QA	1.30	1.68	0.77

Table 3: Between-system spread against within-system seed variation, all runs. A ratio below one means changing the seed moves a system more than changing the system does. All three are below one.

Off-the-shelf XLM-R is not separated from either extended system. It reaches the highest mean on NER and QA while performing no vocabulary extension and no adaptive pretraining, at a fraction of the compute: its runs took 6–24 minutes against roughly 5 hours for the extended systems. We state this as a non-separation rather than a result in its favour; its text-classification standard deviation is the largest in the table, at 15.04.

5 Why the Comparison Does Not Resolve

Table 3 compares the spread between the three systems against the spread between seeds of one system. On every task the second exceeds the first. Excluding the failed runs of §7 does not change the conclusion: the ratios become 0.56, 0.64 and 0.77.

The same point in units of detectable effect: using the observed paired-difference standard deviations, the minimum difference detectable at $\alpha = 0.05$ and 80% power is 19.80 macro-F1 on text classification, 3.10 F1 on NER and 2.32 F1 on QA. The text-classification figure is more than twice the entire observed spread between systems. Reaching a 2-point detectable difference there would require approximately 281 seeds per system. We report this as a property of the benchmark and the sample size, not of any system.

6 A Comparison That Did Not Survive Re-measurement

An earlier batch of runs, under a superseded protocol, reported the morpheme-grounded system ahead of FOCUS on text classification by 1.97 accuracy points. Under the protocol of §3 it trails by 3.32 and wins no seed. We record this because the direction of a published comparison reversed, and because the reason is instructive about comparing across protocol revisions.

The two protocols are not two measurements of the same task. It would be convenient to read FOCUS's movement from 73.80 to 81.62 as the

	Superseded	Current
Test items	71	235
Majority-class share	35.21%	66.38%
Documents in corpus	700	2,340
LGSE (accuracy)	75.77	78.30
FOCUS (accuracy)	73.80	81.62
LGSE − FOCUS	+1.97	−3.32

Table 4: The two text-classification protocols. They differ in test set, training set and class prior together, so the accuracy columns are not two measurements of one quantity and the per-system shifts are not interpretable as effects. What the table shows is that the comparison between the two systems reverses. Per-class supports are in Appendix B.

effect of a larger test set on a system that performs no morpheme segmentation. That reading is not available. Rebuilding the corpus changed the training split and the label distribution as well as the test split: the majority class holds 35.21% of the superseded test set and 66.38% of the current one, so a classifier that predicts only the majority label gains 31.17 accuracy points from the change alone, with no change in capability. Accuracy on the two splits is scaled differently, and no per-system difference between the columns can be attributed to a single cause.

What the reversal does establish. Within the current protocol, measured on one split with one configuration and shared seeds, the morpheme-grounded system does not lead FOCUS on text classification; it trails by 3.32 points and loses every seed. That comparison is internally valid and is the one reported in Table 1. The earlier favourable comparison is superseded, not contradicted by a controlled intervention, and we do not claim to have isolated why it differed.

Consequence for the missing systems. Two systems from the original design were not run under the current protocol (§10). Their numbers exist only in the superseded batch, on the 71-item split with a 35.21% majority class. Placing those values in Table 1 beside systems measured against a 66.38% majority class would produce an ordering driven by the class prior rather than by the systems. We therefore leave those rows empty rather than fill them across protocols.

7 A Fine-Tuning Failure Mode

Six of 60 runs never fit their training data. Their relative training-loss reduction from the first to the last logged epoch falls between 21.7% and 28.9%, against 69.5% to 97.9% for every other run—a gap with no runs inside it. Any threshold between 28.9% and 69.5% selects the same six, so the classification does not depend on where in the gap it is placed. The rule uses training loss only and never a test score.

Task	System	Seed	Loss reduction
NER	FOCUS	44	24.1%
NER	LGSE	45	21.7%
TC	LGSE	42	28.9%
TC	LGSE	43	24.8%
TC	XLM-R	42	21.9%
TC	XLM-R	45	26.8%

Table 5: The six runs selected by the training-loss rule. They occur under all three systems, so the instability belongs to the fine-tuning setup rather than to any initializer.

The resulting predictors are degenerate in a task-specific way. On NER the tagger proposes no entity at all, giving precision, recall and F1 of exactly zero. On text classification the classifier never predicts class 3, which holds 19% of the test set, and never predicts the two rarest classes; per-class F1 shows this as a partial label collapse rather than a total one. Dev scores track test scores in each case, so the failure is visible without touching the test set.

These runs are not checkpoint mis-selection: the evaluation-loss minimum across all ten saved checkpoints is 0.77–0.80 in the failed runs against 0.42–0.55 in the successful ones, so no available checkpoint was good. Nor is it overfitting: training loss stays high.

We retain all six runs in Table 1 and report a secondary analysis excluding them. Excluding them moves the text-classification means to 54.67, 54.56 and 56.45 for LGSE, FOCUS and XLM-R, on three, five and three runs, and the NER means to 68.27, 66.78 and 67.05 on nine, nine and ten runs. Neither policy separates the systems: the intervals remain wide and overlapping under both. We report both because the choice moves the text-classification means by nine points and the NER means by seven, and we see no principled basis for preferring one.

8 Recommendations

Each observation above has a cheap remedy that we would have benefited from.

Check resolution before comparing. Whether the between-system spread exceeds the between-seed spread is computable from a pilot, before any method is proposed. Where it does not, as in three of three tasks here, a five-seed comparison cannot support a ranking regardless of which system wins.

Report the minimum detectable difference. One line per comparison. It distinguishes "no difference was found" from "no difference of this size could have been found".

Run a system that shares everything but the contribution. FOCUS differs from LGSE only in morpheme segmentation, so the pair bounds what the segmentation itself can be worth. A system compared only against an unmodified baseline cannot separate the contribution from everything else that changed alongside it.

Do not compare across protocol revisions. When a corpus is rebuilt, the test set, the training set and the class prior can all move together, as they did here (§6): the majority-class baseline went from 35.21% to 66.38%, which alone rescales accuracy by 31 points. Numbers from before and after such a change are not two measurements of one quantity, and a table that mixes them reports the revision rather than the systems. Re-run the baselines or leave the cells empty.

Release the operator, not only the equation. W in Equation 1 is 2.3 MB. Releasing it, or the code that builds it, removes the largest obstacle to reproducing a method of this kind. We release ours with a sidecar recording its language, anchor count, objective and residual.

Disclose degenerate runs. A single all-○ seed can move a mean by 13 points and inflate a standard deviation past 30. Neither silent exclusion nor silent averaging is adequate; report both and state the rule.

9 Related Work

Initialization for vocabulary extension. WECHSEL (Minixhofer et al., 2022) transfers embeddings through bilingual dictionaries and static vectors. FOCUS (Dobler and de Melo, 2023), our near control, builds each new embedding from overlapping tokens in an auxiliary FastText space. Both FOCUS and the morpheme-grounded method reduce to a whole-token vector when their respective condition—overlap, or a decomposition—is unavailable, which is why they are the right pair to compare and why their difference is bounded by how often that condition fires. Csaki et al. (2023) survey the adaptation pipeline. Our results do not argue against this line of work; they indicate that these benchmarks at these sample sizes cannot adjudicate between its members.

Morphological resources for Ethio-Semitic languages. Amharic is templatic and heavily affixing, which motivates morpheme-aware vocabulary construction. HornMorpho (Gasser, 2011) provides finite-state analysis for Amharic and Tigrinya and is enabled in the configuration used here; with it, 99 of the 198 added Amharic tokens (50.0%) receive a decomposition into more than one morpheme, against 33 (16.7%) from the static lexicon alone. We report these as properties of the pipeline and do not use them to explain the comparison outcomes.

Reproduction and statistical practice. Belz et al. (2021) survey reproducibility in NLP. Our methodology is control-based: rather than reporting that numbers differ, we hold systems that should be unaffected by a change fixed and attribute the difference from their movement.

10 Conclusion

Under one protocol, on three Amharic tasks, no paired comparison between morpheme-grounded initialization, overlap-based initialization and no extension at all excludes zero. The reason is measurable and specific: changing the seed moves a system further than changing the system does, the smallest detectable difference on text classification is twice the observed spread, and a tenth of runs fail to fit their training data. A comparison reported under a superseded protocol reverses under this one.

We do not conclude that these initializers are equivalent; our tests are too small to support that, and we say so. We conclude that this evaluation suite, at the sample sizes the field conventionally uses, does not answer the question it is being asked, and that the four properties above are worth measuring before the next comparison of this kind is published.

Limitations

Our W and λ are documented substitutions, not values recovered from the original method, so the

LGSE column is one faithful instantiation and the absolute numbers would differ under another.

Two systems from the original design—adaptive pretraining alone, and extension with random initialization—were not run under this protocol. An earlier batch contains them, but evaluated text classification on the superseded 71-item test set, carries three different configuration checksums, and mixes two alignment matrices; §6 shows that placing those numbers beside these would create an ordering artifact of about 8 points. We therefore leave the rows empty rather than fill them from a different protocol. Completing them under this protocol would add two rows at five seeds each, which Table 3 and the detectable-difference figures indicate would not change the conclusion.

The paired tests have five to ten observations. They establish that these data do not separate the systems, not that the systems are equivalent; distinguishing those readings would need many more seeds or benchmarks with a larger between-system spread.

This study covers Amharic. No cell for a second language exists under this protocol, so we make no cross-linguistic claim.

The text-classification corpus is small, its label distribution is heavily skewed, and one of its six labels has no test instances. Our primary metric excludes that label; the recorded six-class value is reported alongside for traceability.

Ethics Statement

This paper replaces a retracted paper by the same authors. We report the outcome as measured, including where it contradicts our own earlier claims, and we document errors made during this work alongside those found in the original: a configuration error that applied the Tigrinya alignment matrix to Amharic runs, a morpheme-coverage figure initially over-reported before re-measurement, and three log-parsing defects in our own analysis code, each of which produced a wrong failure classification and is now pinned by a regression test. We found no evidence of fabrication in the original work; every discrepancy we traced is consistent with ordinary pipeline defects of the kind we also made here.

Self-examination carries a conflict of interest. We mitigate it by pre-registering the one directional control before it ran (§6), by releasing every per-run record so that each number can be recomputed independently, and by reporting the outcome least favourable to our own prior claim. Readers should weigh our conclusions accordingly.

All datasets are publicly available or released by us. The Amharic educational-quality annotation was led by the first author. No human subjects were involved.

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A Provenance

One script. Every number in this paper is produced by `scripts/build_paper_tables.py`, which reads the per-run `experiment.json` records and emits Tables 1, 2, 3, 5, the per-class text-classification breakdown, the coverage counts, and a per-seed CSV. Running it reproduces every table; nothing in this paper was typed by hand.

Runs. 60 runs, all from `results_hornmorpho_wam/`: three systems × three tasks × five seeds, with NER extended to ten. Each record carries its commit, configuration checksum, alignment-matrix path, corpus checksum and per-seed dev and test metrics. All 60 share `config_sha256 3e51cdab...`.

Excluded directories. Four other result directories exist and are excluded for protocol differences, never for their outcomes: `results_gen2_snapshot_20260924` and `results` (150 runs each; three configuration checksums, two alignment matrices, 71-item text-classification test set), `results_hornmorpho` (54 runs; mixes the two alignment matrices), and `results_tigrinya_gen2` (45 runs; Tigrinya only, no three-system comparison).

Recipe. Adaptive pretraining: frozen backbone, maximum length 256, MLM probability 0.15, learning rate 5×10^{-5} constant, batch size 32, 10 epochs, weight decay 0.01, AdamW, fp16. Fine-tuning: the same learning rate, schedule, batch size and epoch count, with checkpoint selection by evaluation loss. Identical for all three systems, untuned. FastText: skipgram, dimension 768, 5 epochs, minimum count 5, vocabulary 165,471. Vocabulary extension adds 198 Amharic tokens.

Compute. Each extended-system run takes approximately 5 hours on one A100, dominated by adaptive pretraining, which is repeated per seed because the seed determines the initialization under test. XLM-R runs take 6–24 minutes. The 60 runs total roughly 230 GPU-hours.

Per-seed values. Released as `tables/per_seed.csv`: one row per (task, system, seed) with the score, the failure flag, the commit and the result file path.

B The Two Text-Classification Protocols

Per-class test supports, read from the run records of both batches. The superseded protocol is retained here only to document why its accuracy figures are not comparable with the current ones (§6); no number from it enters Table 1.

Label	1	2	3	4	5	6	total
Superseded	25	24	13	7	0	2	71
Current	25	156	45	7	0	2	235

Table 6: Class 2 grows from 24 to 156 instances, taking the majority share from 35.21% to 66.38%. Label 5 occurs in training but in neither development nor test split under either protocol, which is why the recorded six-class macro-F1 is capped at five sixths of its maximum (§3).